Ifan Hakim

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Informatics fresh graduate specializing in Data & AI, especially in the fields of Machine Learning and LLM Orchestration. 1 year experience in the Data & AI lifecycle for manufacturing and healthcare, equipped with strong project management and business analysis skills. Passionate to applying AI with impactful and measurable solutions.

WORK EXPERIENCE

AI Engineer Cohort (Intensive Bootcamp) — **Feb 2026 - Jun 2026**
DBS CODING CAMP 2026 — **Remote**

- Developed "SugarCoat.ai," an early diabetes risk detection platform integrating Machine Learning classification and a Generative AI health coach, enabling personalized health monitoring through non-medical lifestyle data acquisition.
- Orchestrated the end-to-end deployment of an ML inference API to Hugging Face Spaces utilizing Docker containerization, establishing a scalable and isolated environment for real-time model serving.
- Mastered the end-to-end Machine Learning Engineering lifecycle across 8 core modules, gaining robust proficiency from model development to deployment, including the integration of Large Language Models (LLMs) via the OpenRouter API.

Smart Manufacturing Intern — **Oct 2025 - Apr 2026**
KALBE NUTRITIONALS (PT. SANGHIANG PERKASA) — **Cikampek, Indonesia**

- Engineered an integrated HVAC Digital Twin 2.0 using First Principles and Machine Learning, achieving real-time optimization of HVAC performance with thermodynamic and operational accuracy.
- Architected an automated IoT data pipeline via Node-RED to streamline historical sensor data acquisition, forming a robust foundation for continuous model training and validation.
- Developed a Preventive Maintenance application to digitize inspection checklists, resulting in a 25% increase in operational efficiency.

Pricing Strategist for Solar Windows (Project-Based) — **Dec 2024 - May 2025**
PRISTINZ SOLUTIONS — **Remote**

- Designed pricing strategy for solar window using value-based method and competitor analysis method.
- Calculated projected profit margins within 1–2 years through strategic pricing models.
- Recommended early adopter incentives to enhance adoption and ROI potential.

AI for Medical Device Industry (Independent Study) — **Aug 2024 - Dec 2024**
PT STECHOQ ROBOTIKA INDONESIA — **Sleman, Yogyakarta**

- Built and modified the EfficientNetB0 model for tumor classification, achieving a baseline accuracy of 98% and establishing a standardized experiment reproduction process that improved efficiency by 30%.
- Established a preprocessing and cross-validation pipeline that reduced overfitting by 80% and improved model generalization, resulting in a 18% increase in predictive accuracy.
- Served as Project Manager for the project, managing the timeline, documenting each stage of development, and presenting the final results to stakeholders and mentors, ultimately winning the “Most Collaborative Mentee” award.

EDUCATION

Bachelor of Computer Science (B.C.S) in Informatics — **Sep 2022 - Nov 2026**
Universitas Nahdlatul Ulama Yogyakarta — **GPA:** GPA: 3.95/4.00

- Achieved a high GPA of 3.95/4.00 while pursuing a Bachelor's degree in Informatics with a deep specialization in Artificial Intelligence.
- Graduated with a thesis titled “Design of a Digital Twin for Optimizing Cooling Tower-Condenser Systems Based on the Integration of Machine Learning and First Principles in HVAC Systems”.
- Demonstrated leadership and communication skills by serving as Secretary of the Informatics Student Association and being selected as a Teaching Assistant for various technology subjects.

SKILLS

Tech Stacks: Python, Pandas, NumPy, Matplotlib, Seaborn, Streamlit, Scikit-learn, PyTorch, TensorFlow & Keras, FastAPI, LangChain, Git, GitHub, Docker, Huggingface, Google Colab, Visual Studio Code, Open-Router.

Soft Skills: Project Management, Business Analysis, Collaboration, Leadership, Problem Solving, Time Management, Critical Thinking.

LANGUAGES

Indonesia (Native proficiency) • **English** (Professional working proficiency)

CERTIFICATIONS

English for Business Communication by **USG Education** — **Jun 2026**
Building Advanced Deep Learning Projects (Credential ID: MEPJOG92LZ3V) by **Dicoding** — **Jun 2026**
Learn Fundamental of Deep Learning (Credential ID: MEPJO96LLZ3V) by **Dicoding** — **Mar 2026**
Learn Git Basics with Github (Credential ID: EYX4QRQ45PDL) by **Dicoding** — **Feb 2026**
Started with Python Programming Language (Credential ID: 1OP8RL2YVZQK) by **Dicoding** — **Feb 2026**
Learn Machine Learning for Beginners (Credential ID: 0LZ0Y1N5NX65) by **Dicoding** — **Feb 2026**

Getting Started with the Basics of Programming to Become a Software Developer (Credential ID: 6RPN7KYD9X2M) by **Dicoding** — **Feb 2026**

Learn Artificial Intelligence Basics (Credential ID: L4PQ9MNQOPO1) by **Dicoding** — **Feb 2026**

Learn Data Science Basics (Credential ID: 0LZ0R3LMNP65) by **Dicoding** — **Jan 2025**

EF SET English Certificate 67/100 (C1 Advanced) (Credential ID: r3Ff5k) by **EF SET** — **Dec 2024**

Artificial Intelligence In Industry 4.0 For Medical Device Industry by **Stechoq Academy** — **Dec 2024**

Learn Data Analytics with Python (Credential ID: 81P2NJOGOXOY) by **Dicoding** — **Aug 2024**

Learn Structured Query Language (SQL) Basics (Credential ID: 07Z60M09MZQR) by **Dicoding** — **Jul 2024**

Learn Data Visualization Basics (Credential ID: 98XWLV8W0ZM3) by **Dicoding** — **May 2024**

AWARDS

Excellence Awards: Outstanding Potential in Smart Environment Mapping by **International Conference on Sustainable Innovation and Technology (ICSIT'25)** — **Dec 2025**

Received the Category Excellence Awards: Outstanding Potential in Smart Environment Mapping at the ASEAN Youth Innovation Competition, held in conjunction with the 2nd International Conference on Sustainable Innovation and Technology (ICSIT'25).

Finalist by **Hackvidia by Arkavidia 9.0** — **May 2025**

Being Top 8 out of 70+ teams in the Hackathon Competition by Bandung Technology Institute, Bandung.

The Most Collaborative Mentee by **PT. Stechoq Robotika Indonesia** — **Jan 2025**

In Studi Independen Bersertifikat and completing medical image classification project using Deep Learning model and improving team collaboration efficiency up to 60%.

PUBLICATIONS

Optimizing an Expert System for Diagnosing Depression Disorder Using Case-Based Reasoning on **Jurnal Sains, Nalar, dan Aplikasi Teknologi Informasi (SNATI)** — **Jul 2024**

REFERENCES

Septian Ardiansyah (Project Machinery Leader of Engineering Department at Kalbe Nutritionals (PT. Sanghiang Perkasa)) • septian.ardiansyah@kalbenutritionals.com

ORGANIZATIONAL & VOLUNTEER EXPERIENCE

Secretary — **Jul 2024 - Jun 2025**
Informatics Study Program Student Association — **Sleman, Yogyakarta**

- Managed and maintained all organizational documentation, ensuring accurate record-keeping for activities and partnerships, improved retrieval efficiency by 25%.
- Authored detailed meeting minutes and compiled comprehensive management reports, enhancing strategic decision-making efficiency by 30%.
- Initiated and facilitated communication to secure new partnerships, resulting in a 40% increase in the organization's network.

Mentee Batch 5 — **Sep 2024 - May 2025**
StudentsCatalyst — **Sleman, Yogyakarta**

- Developed a comprehensive, value-based pricing strategy for a real-world client, resulting in a projected pricing efficiency increase of 30%.
- Enhanced career-readiness by completing intensive training modules on professional branding, strategic communication, and interview preparation.
- Facilitated career-preparation programs, enhancing participants' personal development plans and expanding professional networks by over 30 connections.

Coordinator of Moderators — **Oct 2024 - Oct 2024**
PyCon APAC 2024 by Python Foundation — **Sleman, Yogyakarta**

- Led and coordinated an international team of moderators to ensure the smooth execution of over 50 event sessions, resulting in a 95% satisfaction rate from attendees.
- Developed and delivered briefing materials, ensuring 100% of event issues were resolved in real-time for over 200 attendees.
- Maintained high quality and professionalism standards during Q&A sessions, resulting in a 50% increase in participant satisfaction ratings.

PROJECTS

Website of MTSN 1 Aceh Barat Daya — **Sep 2026 - Sep 2026**

- Architected and developed a high-performance web portal utilizing Next.js App Router and Tailwind CSS, optimizing Core Web Vitals (LCP) through static pre-rendering and automated WebP image compression for an ultra-responsive, mobile-first user experience.
- Integrated Headless Payload CMS v3 with Supabase PostgreSQL to build a secure, decoupled content management system, empowering non-technical school administrators to manage content independently while maintaining zero database hosting overhead.
- Automated school administrative services (PTSP) by engineering a digital download center (13+ forms) and dynamic contact forms powered by Next.js Server Actions and Resend API, streamlining communication and resource delivery for over 500 students and 45+ faculty members.

Indonesian Academic RAG Chatbot — **Aug 2026 - Aug 2026**

- Architected a persistent Retrieval-Augmented Generation (RAG) pipeline utilizing ChromaDB as a local vector store and OpenRouter API,

integrating a multi-step Agentic flow for dynamic query translation and semantic search.
- Engineered an optimized data ingestion system by implementing overlapping text chunking (800-character size, 150 overlap) and batch processing to effectively bypass API rate limits during knowledge base vectorization in Indonesia.
- Resolved SDK formatting constraints by developing a custom embedding function to force float encoding and optimized token cost-efficiency through strategic conversation memory management.

Trash Category Classification **Jul 2026 - Jul 2026**
- Designed a triple-model voting ensemble integrating EfficientNetV2B0, ResNet50V2, and ConvNeXt-Tiny, utilizing dynamically assigned class-specific weights to maximize predictive accuracy across diverse and ambiguous datasets.
- Formulated a custom Weighted TTA strategy that strategically prioritized original image integrity over augmented variants, resolved edge-case predictions and boosting overall model confidence.
- Engineered a memory-efficient inference pipeline on Kaggle kernels by enforcing aggressive VRAM clearing protocols and strict garbage collection, entirely preventing Out-of-Memory (OOM) failures during intensive multi-model prediction loops.

Multivariate Multi-Horizon Time Series Forecasting - Crypto Coin **Jun 2026 - Jun 2026**
- Architected a custom Seq2Seq LSTM model with Multi-Head Attention from scratch to predict 24-hour cryptocurrency trajectories, utilizing multivariate rolling statistics and lag features.
- Implemented advanced model training techniques, developing Custom Training Loops and engineering custom network layers to maximize architectural control.
- Outperformed a Baseline LSTM in forecasting volatile price movements, securing a robust Mean Absolute Error (MAE) of 0.0797 on unseen test data.

SugarCoat.ai **Apr 2026 - Jun 2026**
- Implemented a production-ready diabetes risk prediction model utilizing a Deep Cross Network (DCN) via TensorFlow Functional API and deployed it through HuggingFace Space.
- Integrated custom layer to mathematically penalize high-risk behavioral synergies and deployed the model as a stateless .keras artifact for seamless, low-latency API inference by 30%.
- Boosted lower user friction and driving Gen-Z engagement toward real-world medical lab testing by bridged the DCN backend with a dynamic Generative AI pipeline.

Roblox's Sentiment Analysis with Deep Learning Model **Apr 2026 - Apr 2026**
- Engineered a Deep Learning NLP model (Bi-LSTM/GRU) to classify user sentiments, utilizing the VADER Lexicon for objective ground-truth labeling to bypass biased star ratings.
- Architected the end-to-end pipeline, implementing text preprocessing, tokenization, and overfitting mitigation strategies.
- Achieved a peak testing accuracy of 85.98% through rigorous architectural benchmarking and data-split experiments.

HVAC Digital Twin 2.0 & Smart Maintenance System **Mar 2026 - Mar 2026**
- Architected an end-to-end Digital Twin by consolidating Water-Side (CT-CWP-CHWP) and Air-Side (AHU) modules, creating a unified platform for holistic plant energy analysis and simulation using FastAPI, increasing saving estimation by 20%.
- Deployed an automated ETL data pipeline using Node-RED to capture, clean, and structure historical BMS data, establishing a reliable foundation for continuous model validation.
- Developed a Predictive Maintenance Checklist application using Mendix, digitizing manual workflows to track equipment health anomalies and streamline facility management operations.

AHU & Evaporator Performance Optimization **Dec 2025 - Feb 2026**
- Engineered a high-precision predictive model for Series-Configuration AHUs (Pre & Post-Cooling), utilizing Time-Series Feature Engineering (Lag & Rolling Window) to account for thermal inertia.
- Designed a custom validation pipeline, cross-referencing AI predictions with Enthalpy and Humidity Ratio calculations to ensure physics-compliant outputs, increasing decision-making efficiency by 30%.
- Implemented complex dual-coil load distributions, enabling precise capacity planning and temperature control for variable room demands.

Cooling Tower & Condenser Performance Optimization **Oct 2025 - Dec 2025**
- Developed a First Principles-Machine Learning model using XGBoost to predict Cooling Tower and Condenser performance, integrating thermodynamic formulas to calculate real-time cooling loads.
- Implemented a simulation system under static data constraint and calculated saving estimation from applying configuration reccomendations around 17% in 3 days.
- Achieved optimal energy efficiency recommendations while strictly adhering to operational constraints, such as Approach Temperature and Wet Bulb variations.

Stylomate **May 2025 - May 2025**
- Research and integrating VitonHD model API into MVP application and has been deployed as a Generative AI-powered camera for Virtual Try-On
- Leading a team in designing the "StyloMate" as Hustler and Hacker from concept to final presentation during a high-pressure 24-hour competition.
- Developing a business model with revenue projections of up to IDR 1 billion per year and managed to impressing the judges with our innovation.

Brain Tumor Image Classification **Dec 2024 - Dec 2024**
- Built and modified the EfficientNetB0 model for tumor classification: achieved a baseline accuracy of 98% and established an experiment reproduction process.
- Established a preprocessing and cross-validation pipeline that reduced overfitting and improved model generalization.
- Apply tuning techniques and perform model inference independently using Flask to ensure the quality of the model before it is put into use.